

Intensity-Based Block Matching Algorithm for Mosaicing Sonar Images

Cyril Chailloux, Jean-Marc Le Caillec, *Senior Member, IEEE*, Didier Guériot, *Senior Member, IEEE*, and Benoit Zerr, *Member, IEEE*

Abstract—Registering sonar images to correctly describe seafloors and explain wide geological or biological phenomena is often achieved manually requiring significant human resources. This paper proposes an automatic intensity-based registration algorithm that relies on the optimization of a new similarity measure (SM), within a multiresolution block matching framework. Indeed, several SMs have been evaluated and ranked on real sidescan sonar data to determine the most relevant intensity dependencies between images for matching purposes. Correlation ratio (CR) and mutual information (MI) are then selected and because of their complementary behaviors, merged in a new SM (MI&CR), which performs better than CR or MI alone, to determine robust matching blocks between images. Thus, the proposed two-step registration algorithm uses MI&CR to match two sonar images: a single rigid translation globally matches the images, then a field of locally applied translations is computed for adjusting the final registration to remaining local distortions. Actual processing time can then be tuned according to the required registration accuracy. Due to a survey standard operating mode, only same-survey overlapping images are considered as candidates for matching. Moreover, building mosaics from registered images assumes a flat sea bottom as no global elevation information is provided by sidescan sonar images.

Index Terms—Correlation ratio (CR), mosaic, mutual information (MI), registration, saliency points, sidescan sonar images, similarity measures (SMs).

I. INTRODUCTION

WHILE sidescan surveys produce large amounts of images representing terabytes of data, no automatic registration method exists. Manual identification and matching of landmarks such as rocks or constellation of rocks appearing in the images is the common way to perform registration [1]. Indeed such symbolic methods are the most intuitive as they

consist in a two-step approach: first, a segmentation process associated with a classification step to detect and label specific features, and second, a matching task between these extracted features. Based on this approach, several registration algorithms have been developed [2]–[6]. The main drawbacks of these methods are the time required by the segmentation processing and particularly the fact that feature-based techniques also depend on both the capability and opportunity to correctly detect and identify reliable characteristics (necessary for the matching) on noisy sonar images [7]. An alternative solution is to use iconic (or nonsymbolic) methods based on low level information (such as pixel intensity) compared to symbolic methods that relies on high level features (using sophisticated patterns such as edge or shape descriptor, etc.). An iconic method involves establishing a global relationship between the pixels of two images using only their gray levels [8]. Compared to symbolic methods, iconic algorithms have a main advantage because they tolerate up to a point, either a varying seabed including distortions, illumination variations, occultations, or simply noise contributions on sonar images. First, iconic matching methods have been developed in the middle of the 1970s, using a criterion based on the sum of square differences of gray level intensity. They consider that images are bent by a basic geometric transformation, or corrupted by an additional white Gaussian noise [9] and use either optical flow [10] or phase correlation [11]. From 1980s to today, an increasing number of types of sensors and data appears, in particular in medical [12] or satellite [13] imagery. In those cases, stochastic approaches using a similarity measure (SM) [as mutual information (MI)] have been shown to perform better [14]. In this paper, we present an algorithm dedicated to automatic image registration for sonar images using iconic methods.

To fully describe our procedure, seven main sections are developed. Section II presents several problems encountered when processing multiple sidescan sonar images (occlusions, illumination variations, etc.) and their corresponding solutions are developed. Preprocessing operations such as normalization or projection may be used to reduce this spatial variability. In Section III, various possible approaches for defining the registration algorithm are discussed, depending on the nature of the geometric transformation to be found (rigid or elastic) and on the chosen model of data to be matched (either stochastic or deterministic). Since real phenomena may be more easily described using stochastic models, matching two images relies on the definition of an SM. This section also introduces a methodology to compare various kinds of SMs to estimate their performances for sonar images matching. Thus, Section IV

Manuscript received July 16, 2009; revised June 25, 2010; accepted April 04, 2011. Date of publication August 12, 2011; date of current version October 21, 2011.

Associate Editor: D. Simons.

C. Chailloux was with the Institut TELECOM, TELECOM Bretagne, UMR CNRS 3192 Lab-STICC, Technopôle Brest-Iroise, 29238 Brest Cedex 3, France. He is now with the ENSTA Bretagne, 29806 Brest cedex 9, France (e-mail: cyril.chailloux@telecom-bretagne.eu).

J.-M. Le Caillec and D. Guériot are with the Institut TELECOM, TELECOM Bretagne, UMR CNRS 3192 Lab-STICC, Technopôle Brest-Iroise, 29238 Brest Cedex 3, France (e-mail: jean-marc.lecaillec@telecom-bretagne.eu; didier-gueriot@telecom-bretagne.eu).

B. Zerr is with DGA/GESMA, 29240 Brest Naval, France and also with ENSTA Bretagne, 29806 Brest cedex 9, France (e-mail: benoit.zerr@enseta.fr).

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Digital Object Identifier 10.1109/JOE.2011.2141850

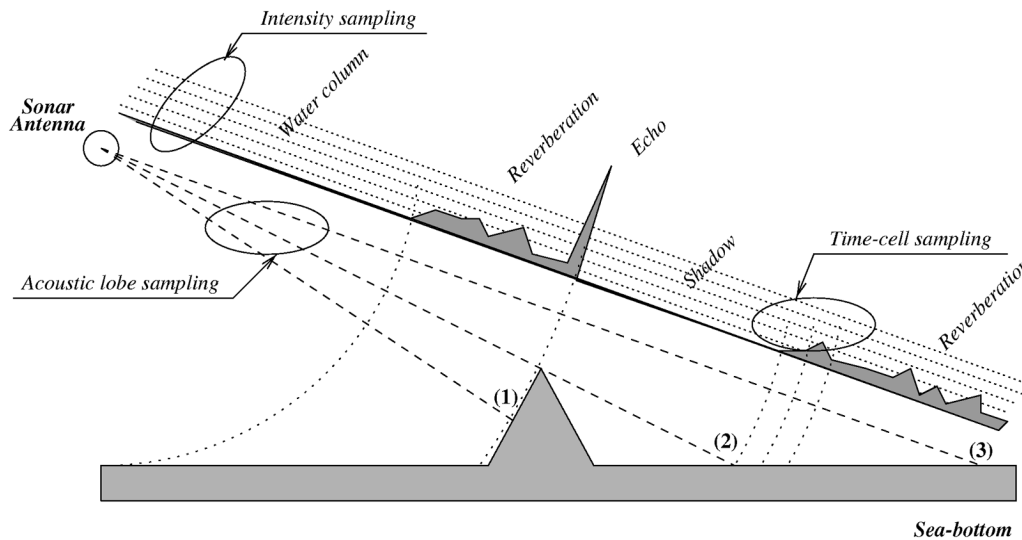


Fig. 1. Building a sonar image line from the backscattered acoustic signal.

is dedicated to the two SMs used in this study: correlation ratio (CR) and MI. Indeed, as these two measures belong to two different families of intensity dependency relationships, the advantages of coupling MI and CR into a new SM will be particularly developed.

Embedding this new SM, Section V describes the full two-step matching algorithm. First, a rigid step computes a global geometric transformation between the overlapping images to match, using a selection of points of interest, through a multi-resolution framework. Then, an elastic procedure operates on the images crudely registered by the global transformation to find the remaining local distortions. In that case, a dense field of locally applied geometric transformations is obtained. Both procedures rely on the coupled MI&CR SM to estimate the best geometrical transformations matching points and their neighborhoods belonging to the two sonar images. Finally, Section VI presents several results obtained with the proposed matching algorithm working on real sonar images. These results are analyzed in the light of their survey-specific acquisition conditions, while Section VII discussion broadens the application of the proposed algorithm to other sonar data sets and configurations with hints about reliability.

II. PROBLEM STATEMENTS

A. Registration and Variability

Matching two sonar images consists in defining the geometrical relationship between features appearing in both images. The key idea is that two matched features correspond to a unique location in the observed environment (or seabed). The meaning of *feature* ranges from individual pixels, contours, and homogeneous textures regions to more complex structures. The human brain is very tolerant when finding the correspondences, especially in terms of accepted deformations and intensity variations between the matched features. Thus, the ultimate goal of our work is to design an algorithm able to automatically perform this task as well as a human operator within a reasonable processing time.

Differences between two collected images may be explained through three families of variability.

- The *modal* variability results from using various types of acquisition sensors (optic, acoustic, etc.). For instance, some underwater systems have both a backward-looking (DUBM-60) and a side-looking (DUBM-42) sonar in order for the first one (called gap filler) to cover the blind nadir area of the second one [15].
- The *temporal* variability is related to the evolution of the observed scene between the two acquisitions, including long-term matching [16].
- The *spatial* variability comes from a change of attitude or viewpoint between the two acquisitions with the same sensor. Dealing with spatial variability includes solving problems such as occlusions [17], illumination [18], orientation, and scale variations.

As our work focuses on sidescan sonar images registration for the same survey (reduced time frame), all the changes between the images to match are related to the spatial variability family. Section II-B will present the issues raised by such changes.

B. The Survey: A Typical Context for Acquiring Sidescan Sonar Images

1) *Sonar Image Construction:* A sidescan sonar usually follows a linear path, sends acoustic pulses, and records the backscattered signal along time, simultaneously on port and starboard sides [19]. For each pulse, beamforming produces one or several beams from the collected signal. As shown in Fig. 1, time-sampling the backscattered signal intensity along a beam allows building one line of the sonar image. The amount of pixels per line depends on the sampling frequency and the recording duration (3). On each line, echoes (1) result from aggregating many returns over a reduced amount of time samples (or pixels) while shadows (2) occur when no backscattered signal is recorded due to relief variation, from (1) to (2). As the sonar moves along its track, new beams are sampled and added to the current image. Thus, the *Y*-axis of a sonar image corresponds to the sonar towfish heading while the *Y*-axis is

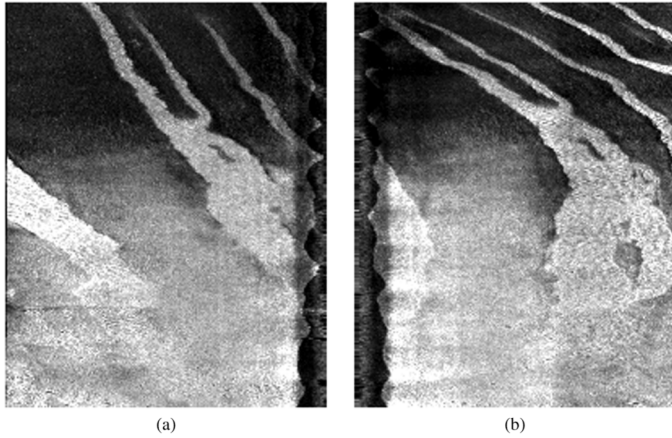


Fig. 2. Sonar data set A (198×250 pixels images of sand and mud). (a) Image A_1 (from track 1—starboard); (b) image A_2 (from track 2—port).

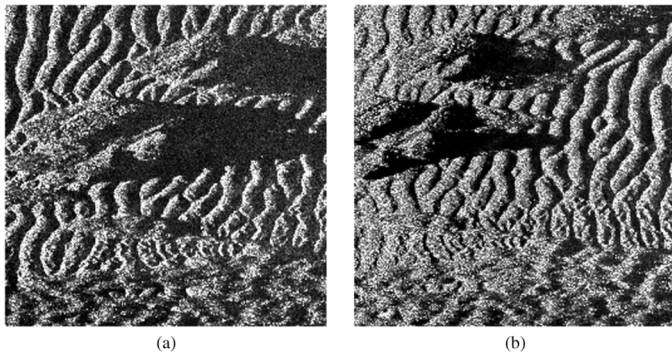


Fig. 3. Sonar data set B (256×256 pixels images with ripples, rocks, shadows, and Posidonia field). (a) Image B_1 (from track 1—starboard); (b) image B_2 (from track 2—port).

perpendicular to X , either clockwise for starboard image or counterclockwise for port image.

Figs. 2–4 present three pairs of sonar images (A, B, and C), extracted from two parallel tracks, on three types of seabeds. These images are collected by a Klein 5500 system in high-resolution mode, with the following predefined parameters: 75-m range with a typical towfish altitude ranging from 10 to 15 m, around 80 m of layback, a $50\text{-}\mu\text{s}$ pulse length giving 3.75-cm across-track pixel resolution, 475-kHz frequency, and 8-kn speed. All the collected images are time-varying gain (TVG) normalized [20] by the acquisition system. Image pair A consists of subsampled full range tracks recorded in the bay of Douarnenez (France) while image pairs B and C are full resolution subimages extracted from midrange sonar data collected in the Ligurian Sea (Italy).

2) *Survey Specificities*: A survey defines the geometry of the trajectories followed by the towfish carrying the sonar system. Typically, a survey is made of tracks, each track covering a small area of sea bottom along a supposedly straight trajectory. Within a survey, these tracks follow a lawn mower pattern. As shown in Fig. 5, the proper acquisition of the seabed area located below the sonar requires a redundant coverage of about 40%, which can be used for registration [21].

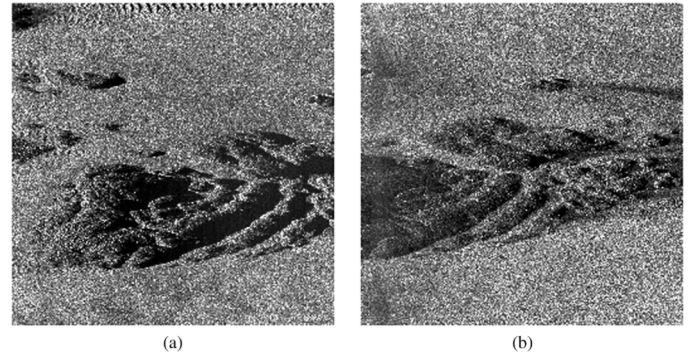


Fig. 4. Sonar data set C (256×256 pixels images of sand, rocks, and shadows). (a) Image C_1 (from track 1—starboard); (b) image C_2 (from track 2—port).

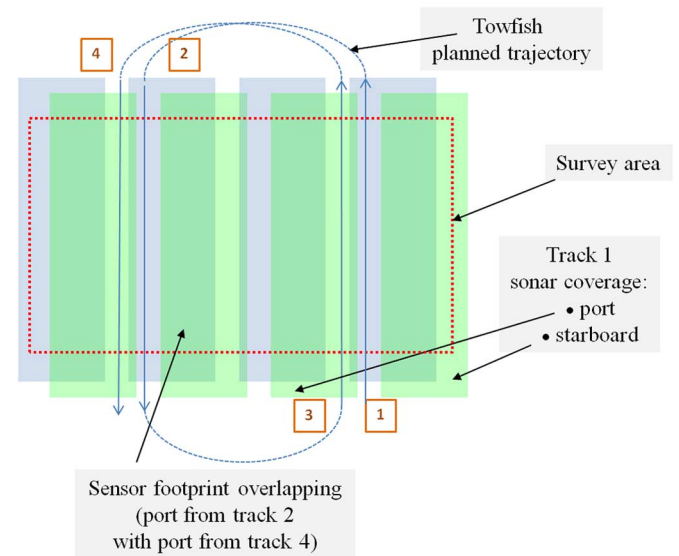


Fig. 5. Typical survey planning with opposite heading and parallel tracks.

C. Impact of Spatial Variability on Sidescan Sonar Images

As presented previously, performing a survey induces some *a priori* knowledge about the collected tracks. This section will take all that knowledge into account to describe various problems facing an automatic algorithm when matching sidescan sonar images.

1) *Illuminations Variations*: The backscattered energy collected by the sidescan sonar decreases as the range increases, because of the acoustic wave absorption: sidescan images appear as *overexposed* at the beginning of the range while very often clearly *underexposed* by the end of the range. Thus, the most appropriate area to perform matching falls around midrange, where illuminations are equivalent.

2) *Occlusions and Shadows*: To overcome the lack of information (in terms of gray level) occurring with shadows or occlusions, the involved areas have been excluded from the registration process by computing a binary mask.

3) *Orientation and Scale Variations*: In the proposed framework, we suppose that the altitude of the towfish remains constant during the survey and the ship cruises at steady speed. The towfish being linked to the survey ship by a cable (layback) may be subject to phenomena (underwater current, etc.) introducing

strong variations in its navigation (both location and attitude). Thus, we will consider that its effective heading may vary within a 90° interval.

4) *Common Data Preprocessing*: Some preprocessing may be applied to the raw sonar images to take into account their acquisition conditions [1]. Image *normalization* tries to compensate for backscattered signal strength alleviation along the range: for instance, a TVG compensation usually forces pixels at a same range to behave according to predefined intensity mean and variance.

Previous works show that sonar filtering may lead to serious losses of information [22], especially when getting rid of speckle. As a consequence, in this paper, the registration algorithm will operate on subsampled and TVG system compensated images to consider these images as raw as possible.

III. CHARACTERIZATION OF THE PROPOSED MATCHING PROCEDURE

Previous sections show that matching sonar images acquired over same sea bottom regions is a crucial procedure. Some questions remain: How to model the matching underlying geometrical transformations? How to compute such geometrical transformations?

A. What Kind of Geometric Transformation: Elastic or Rigid?

In the latter, $\mathcal{T}$ represents the operator mapping a reference image I_X onto a test image I_Y such that I_Y and $\mathcal{T}(I_X)$ match. This mapping operator can be defined through a *rigid* globally applied transformation combining one rotation and one translation. However, as previously mentioned, the layback behavior associated with local acquisitions conditions such as currents clearly involves sonar image matching that will be better performed through an *elastic* transformation. In that case, specific transformations may affect each pixel from I_X to be linked to pixel(s) from I_Y . A simple solution to describe an elastic transformation may be then to consider it as the result of several locally applied rigid transformations [3]. Generally, multimodal elastic matching (the most general case) can be split into a multimodal rigid matching followed by a mono-modal elastic matching.

B. Computing the Transformation: Deterministic or Stochastic Approach?

Matching based on *deterministic* methods operates in a systematic way, generating a local matching on every pixel, like optical flow methods [10], while methods with *stochastic* models involve less constraining hypotheses on the intensity, considering sonar images as random variable sets. Preferring stochastic methods is also motivated by a preliminary study [22], which suggested that deterministic methods are not able to manage sidescan image matching, because of strong illumination variations and too large displacements, even within a multiresolution frame.

To evaluate and compute relevant geometrical transformations, one of the stochastic models key features is to define a distance, namely an SM, between two sets of random variables trials. Thanks to this measure, comparisons between subimages can be done and the best mapping between images found.

Moreover, through this approach, images coming from different modalities can be dealt with, even with strong distortions or illumination variations.

C. Main Features of the Proposed Algorithm

1) *A Block Matching Approach*: To perform elastic matching, parts of image must be registered rather independently. Classical *block matching* approaches are appropriate as they perform intensity matching between subregions (blocks) from the two images to match, all over the two images. To only match relevant blocks from the two images, hybrid algorithms [23] only perform block matching around detected features in image I_X , reducing the amount of comparisons to be done. Moreover, they allow one-view specific parts of an image not to participate in the whole matching procedure, and local large transformations to be estimated, bringing extra robustness when dealing with shadows, occlusion, and layback perturbations. This split and focus approach is rather different from global approaches such as Fourier transform matching [24], which work over the full image injudiciously taking into account nonredundant features.

Thus, we choose a hybrid two-step method to characterize the matching operator $\mathcal{T}$: the first step which extracts maximum curvature points as feature points for a *block matching* algorithm, followed by local SM estimations. Transformation $\mathcal{T}$ becomes a sparse geometric transformation $\mathcal{T}_s$ matching all the defined blocks in image I_X with the best blocks in image I_Y . Focusing only on points of interest for computing a relevant local matching heavily decreases the required processing time.

2) *A Multiresolution Framework*: A multiresolution algorithm is necessary to deal with variable scale transformations and compensate for large displacements. Indeed, laybacks introduce both small and large distortions between the images. Our multiresolution approach also consists in exploiting the result of each resolution level independently to produce a full vector transformation field, as described in Section V-B2.

3) *A Symmetric Matching Operator*: The proposed matching algorithm has to be symmetric as none of the two images to match (test and reference images) really represents reality. Selecting points of interest only on the reference image introduces a bias. To overcome this, our matching algorithm must operate twice, switching the *test* and *reference* images. Thus, the final result may integrate both images influence, merging contributions from both transformation vector fields.

4) *Which SM to Consider?*: As previously said, the crucial feature of a stochastic modeling algorithm is to compare two sets (or blocks) of random variables trials (pixel intensity values). Many specific SMs can be defined but they belong to four families according to the assumption that they make about the kind of a relationship between the intensities of the two blocks to check. Some SMs assume a linear relationship between the two-block pixel values [25]; others propose a second-order dependency between pixels [25] or even a more general relationship through functional dependencies [26]. Finally, a law dependency based on the probability density function properties of the two blocks can also be exhibited [27].

About this topic, one major contribution of this paper is the selection and the fusion of two SMs among the numerous possi-

TABLE I
SMS PERFORMANCES ACCORDING TO INTERINTENSITY DEPENDENCY RELATIONSHIP

Types of dependency	Similarity measures	Set A		Set B		Set C		Rank
		ρ_1	ρ_2	ρ_1	ρ_2	ρ_1	ρ_2	
Second order dependency [25]	Correlation	0.029	608	0.039	872	0.035	763	36
	Sum of Absolute Differences (SAD)	0.142	128	0.447	73	0.054	519	28
	Sum of Squared Differences (SSD)	0.214	225	0.033	410	0.327	501	29
	Correlation coefficient of Moravec	0.773	254	3.005	113	2.09	16	8
	Median of grey level differences	0.197	381	0.413	158	1.52	65	21
	Least median of squared differences	0.216	225	0.033	410	0.329	501	27
	Seitz measure	0.169	298	0.125	289	0.048	712	32
	Variance of Squared Differences (VOSD)	0.102	372	0.045	963	0.03	404	35
	Kurtosis	0.219	225	0.034	410	0.331	501	25
Linear dependency [25]	Zero Normalized Cross Correlation (ZNCC)	0.744	353	3.106	116	2.05	15	10
	Normalized Covariance (NCOV)	0.079	849	0.138	326	0.111	19	33
	Cross correlation coefficient	0.021	305	2.727	32	0.028	575	22
	Zero Normalized Sum of Squared Differences (ZNSSD)	0.651	235	1.321	144	1.44	41	13
	Smooth Median Absolute Deviation (SMAD)	0.465	68	0.058	304	7.07	70	16
	M-estimator (Geman MacClure)	0.267	93	0.065	505	1.61	606	20
	Pseudo-norm ($\alpha = 0.5$)	0.177	345	0.444	264	0.068	607	23
Functional dependency [26]	Phase correlation [28]	0.769	143	0.367	153	1.08	23	17
	Woods criterion	0.652	289	1.481	149	0.878	531	14
	Correlation ratio	0.490	203	2.711	41	2.93	21	11
Law dependency [27]	Dissimilarity of χ^2	0.162	519	0.158	82	0.024	626	31
	Distance to independence	0.136	70	0.476	128	7.9	9	18
	Mutual Information	1.315	41	2.668	44	15.04	9	1
	Kolmogorov distance	1.428	40	2.188	58	8.78	10.5	3
	Kullback Leibler divergence	0.389	78	2.553	53	16.04	13.3	6
	K-divergence of Lin	1.872	76	2.294	59	7.96	7	2
	L-divergence of Lin	0.396	78	2.550	35	15.8	11	7
	Hellinger distance	1.841	18	2.207	65	7.89	7	4
	Toussaint distance	1.369	40	2.539	36	3.16	8	9
	W-divergence of Kagan	0.148	35	0.625	143	2.13	4	19
	Matusita distance	1.405	41	1.646	59	8.81	9.6	5
	Battacharya coefficient	0.154	607	0.150	85	0.19	223	26
	χ^2 divergence of Vajda	0.241	300	0.0182	155	0.093	291	30
	α order information of Bose Einstein	0.249	147	2.144	59	1.77	18.6	15
	α order information of Rényi ($\alpha = 3$)	0.325	431	0.035	744	0.132	317	24
	α order information of Femi Dirac	0.098	703	0.036	442	0.059	510	34
	Cluster Reward Algorithm (CRA) [13]	0.428	147	1.867	68	6.7	245	12

bilities that have been tested. Table I proposes a synthetic review of their individual performances. The next section will describe the whole procedure, starting from the definition of the criteria used to select the best SMs to be merged later on.

Moreover, as SMs are only estimated for relevant image areas (the blocks are centered on points of interest), the whole matching process will not be disturbed by occlusions and shadows.

IV. MI&CR: A NEW SIMILARITY MEASURE

A. Identifying Relevant SMs

To estimate the quality of any registration algorithm, a supervised (manual) registration is considered as the goal to achieve, and used to measure performances criteria with respect to this optimal registration.

Most of the time, authors compare SM performances using synthetic data corresponding to a regular grid distorted by an affine transformation and corrupted with predefined noise, thus

discarding any random elastic transformation found in real data. To benchmark the tested SMs, we choose to use real sonar data sets acquired during several different surveys, from Figs. 2–4.

1) *Methodology for Estimating Geometric Transformations:* Estimating the performances of an SM relies on evaluating how close is the resulting geometrical transformation from the effective one, given by a human expert.

To do so, the first ten points having the highest curvature [29] estimated on the image are selected. For a given point i (of interest), all the SMs between this point and all the points in the test image are estimated over their neighborhood (i.e., a block). Finally, the pair (point i in the reference image, point in the test image) with the highest similarity measure estimate (SME) s_i will define the geometric transformation associated with the point of interest i . Comparing the values s_i coming from different SMEs requires a rescaling so that all the values from each individual SME range from 0 to 1.

2) *Definition of Performance Criteria:* As a performance indicator, for each point of interest i , the distance d_i (in pixels) between the expert geometrical transformation and the estimated

one defines the matching error. Based on these data, two criteria ρ_1 and ρ_2 are defined.

- a) *SME versus matching error*: The first performance criterion ρ_1 sums the ratios between the normalized SME s_i and the matching error d_i , over the ten points of interest

$$\rho_1 = \sum_{i=1}^{10} \frac{s_i}{\max(d_i, \varepsilon)}. \quad (1)$$

According to ρ_1 , the best matching is logically obtained when ρ_1 maximum is reached (s_i high, d_i low). However, to deal with perfect matching ($d_i = 0$, for all i), the d_i lower bound is set to $\varepsilon = 10^{-1}$ to grant perfect matching ratios higher than for nonperfect ones.

- b) *Matching errors accumulation*: A second criterion ρ_2 sums the matching errors d_i corresponding the stronger SMEs s_i , i.e., above the median value 0.5. This thresholding makes this criterion more robust, discarding questionable matching. Thus, low values for ρ_2 indicate small matching errors

$$\rho_2 = \sum_{i=1}^{10} \mathbb{1}_{\{s_i \geq 0.5\}} \cdot d_i \quad \text{with} \quad \mathbb{1}_x = \begin{cases} 1, & \text{if } x \text{ true} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

3) *Results*: Table I gathers all the results obtained with more than 35 SMs on the three sonar data sets from Figs. 2–4. The last column of the table contains their rank according to their matching achievement. This rank is established by sorting the ρ_1 average values over the three data sets, ρ_2 criterion making the difference in case of similar ρ_1 .

The ranking given in Table I is consistent with the performances found by Sarrut [30]. Moreover, among the four kinds of dependency that have been studied, we observed that the linear dependency and the second-order dependency are unproductive on sonar images [22]. In any case, a linear relationship hypothesis between intensities is no more valid when several modalities are involved as discussed again in Section VII.

In the latter, we focus on the most robust ones describing either functional or law dependencies. These two SMs are:

- the CR that reflects a functional dependency;
- the MI modeling a dependency in law.

Most of the measures characterizing law dependencies are quite equivalent because they come from the Havrda–Chavrat entropy and can be computed from one to another through a linear combination [27], [31]. It explains why we only choose, among all the measures of dependency in law, the MI measure. The second measure (CR) is clearly not the best ranked, but as it belongs to another family (functional dependency), it may prevent from combining redundant information.

4) *Towards a Fusion Approach*: For this analysis, sonar data are not filtered (except for TVG filtering and with the particularity of data set A which has been subsampled) so that MI measure is not so penalized because of possible loss of information. However, MI weaknesses still remain with difficulties to accurately localize maximum peak of similarity [32], to represent spatial pixels organizations [33], and finally, to consistently process data in a varying scale framework [34]. Some authors propose solutions to overcome these problems, allowing

MI to integrate spatial information. Among them, we can name regional MI, gray level co-occurrences matrices (GLCM), Haralick image parameters, and local variance measures. All these solutions turned out to be inefficient on our sonar images [22].

Thus, an extrinsic solution to MI has been considered, building a new SM: MI&CR. Indeed, the key idea is to couple MI with CR measure to better integrate pixel spatial information as the CR measure has proved to perform better in multiscale frame matching [35], [36] and particularly when information level is weak, in low resolution (64×64 pixels images). On the contrary, MI performs well for high-resolution images. This shows the interest of combining these two SMs into a new one. The next sections will deeply describe the coupling procedure.

B. CR and Functional Dependency

This kind of dependency occurs when one function, possibly nonlinear, links all the pixel intensities from the two blocks X and Y to match. The CR SM reflects this kind of dependency [26] as it consists in minimizing the intensity conditional variance, normalized by the intensity variance. Such a measure is well suited to explain X according to Y . Knowing Y in relation to X permits to use the theorem of total variance

$$\text{var}\{Y\} = \text{var}\{\mathbb{E}\{Y|X\}\} + \text{var}\{Y - \mathbb{E}\{Y|X\}\} \quad (3)$$

where $\mathbb{E}\{Y|X\}$ is the conditional expectation of Y knowing X .

Equation (3) appears as an energy conservation law, with two terms of antinomic energy: while $\text{var}\{\mathbb{E}\{Y|X\}\}$ measures how Y is predicted by X , $\text{var}\{Y - \mathbb{E}\{Y|X\}\}$ measures how Y is functionally independent of X . From (3), it is important to note that $\text{var}\{Y - \mathbb{E}\{Y|X\}\}$ can be weak for two different reasons:

- Y is *well explained* by X (i.e., $\text{var}\{\mathbb{E}\{Y|X\}\}$ is close to $\text{var}\{Y\}$);
- Y contains few information (i.e., weak $\text{var}\{Y\}$).

Thus, to get the best $\text{var}\{\mathbb{E}\{Y|X\}\}$ (i.e., close to $\text{var}\{Y\}$), one solution is to compare the *well-explained* energy part of Y with the total energy $\text{var}\{Y\}$, by minimizing a so-called CR $\text{CR}(X, Y)$

$$\begin{aligned} \text{CR}(X, Y) &= \frac{\text{var}\{\mathbb{E}\{Y|X\}\}}{\text{var}\{Y\}} \\ &= 1 - \frac{\text{var}\{Y - \mathbb{E}\{Y|X\}\}}{\text{var}\{Y\}}. \end{aligned} \quad (4)$$

This ratio takes values from 0 when there is no functional dependency, to 1 when all pixels share the same dependency. As X and Y may not have the same distribution law, most of the time $\text{CR}(X, Y)$ is different from $\text{CR}(Y, X)$. This nonsymmetry will be taken into account when performing the full algorithm by switching reference and test images, as shown in Section V-B2.

C. Mutual Information and Dependency in Law

Considering the two blocks to be matched as two sets of random variables X and Y , a powerful approach is to measure the distance between their joint law $f_{X,Y}(x, y)$ and the product $f_X(x) \cdot f_Y(y)$ of their marginal laws. Indeed, the joint histogram of two independent images that estimates their joint law will only consist of clusters of sparse points. On the contrary, the joint histogram of proportional images reveals an affine line.

Hill shows that the joint histogram dispersion is closely related to the strength of image dissimilarities [37].

Information theory has also developed interesting measures such as joint entropy [12] or MI [14] to quantify joint histogram dispersion. Indeed, as the entropy $H(X)$ measures the quantity of information that an event X produces, the less an event is supposed to occur (i.e., $-\ln(f_X(x))$ high), the more its associated information is important

$$H(X) = - \int f_X(x) \ln(f_X(x)) dx. \quad (5)$$

Considering two images, marginal and joint entropies can be defined to integrate cross behaviors

$$H(X, Y) = - \int \int f_{X,Y}(x, y) \ln(f_{X,Y}(x, y)) dx dy. \quad (6)$$

Using these information measures on two images X and Y , MI computes the relative entropy between their joint law $f_{X,Y}(x, y)$ and the product $f_X(x) \cdot f_Y(y)$ of their marginal laws as defined by

$$\begin{aligned} MI(X, Y) &= H(X) + H(Y) - H(X, Y) \\ &= \int \int f_{X,Y}(x, y) \ln \frac{f_{X,Y}(x, y)}{f_X(x) f_Y(y)} dx dy. \end{aligned} \quad (7)$$

As a consequence, after normalization, MI takes values ranging from 1 when the law matching is optimal, to 0 if no similarity is involved.

With (7), MI weakness on spatial information description is obvious. We suppose that CR, which performs better than MI in a local framework, can compensate for this weakness. That is the reason why we propose a new SM coupling MI and CR within a multiresolution frame to emphasize their complementarities.

D. Coupling CR With MI for a New SM

By embedding intensity variance computations, the CR measure quantifies relationships of proximity between pixels, not described by the MI measure. On the other hand, MI allows catching more complex dependencies. Moreover, when working with low-resolution images, CR appears to be much more relevant than MI (see Table II). Several authors have also observed that variance behaves better than entropy on small data samples [38], [39]. Thus, building a new SM merging CR and MI will take into account their complementary behaviors.

1) *Defining MI&CR, the New SM:* Using MI and CR jointly has already been applied for infra-red satellite classification in order to either enhance statistical dependencies between classes [40] or match two multi-modal images mixing the measures into a criterion to be minimized [41].

In this paper, we propose a new SM, MI&CR, that benefits from MI and CR properties to characterize sonar images X and Y . Equation (8) defines MI&CR as the MI and CR geometrical mean. Geometrical mean has been preferred to arithmetical mean as merging operator, because it appeared that the product brings more stability than the sum, when tested on A, B, and C sonar data sets

$$MI \& CR(X, Y) = CR(X, Y) \cdot \sqrt{1 - \exp(-2MI(X, Y))}. \quad (8)$$

TABLE II
ESTIMATING THE RELEVANCY OF COUPLING MI AND CR SMS FOR MATCHING DATA SET B RAW SONAR IMAGES USING VARIOUS LEVELS OF RESOLUTION. THE EFFECTIVE TRANSLATION TO BE FOUND FROM IMAGE B_1 TO IMAGE B_2 IS $(\Delta x, \Delta y) = (-65, -4)$ PIXELS

Resolution 64×64			
	β	Δx	Δy
MI _{1→2}	1/19	-48	2
MI _{2→1}	0/19	-	-
CR _{1→2}	3/19	-68.64	2
CR _{2→1}	5/19	28.8	-15.6
MI & CR _{1→2}	6/19	-67	4
MI & CR _{2→1}	9/19	30.44	-12
Resolution 128×128			
	β	Δx	Δy
MI _{1→2}	2/19	-79	-9
MI _{2→1}	9/19	72.8	3.32
CR _{1→2}	6/19	-66.16	4.32
CR _{2→1}	8/19	71.74	2.5
MI & CR _{1→2}	7/19	-66.56	2.84
MI & CR _{2→1}	8/19	71.74	2.5
Resolution 256×256			
	β	Δx	Δy
MI _{1→2}	11/19	-65.36	5.81
MI _{2→1}	14/19	76.28	3.85
CR _{1→2}	10/19	-63.7	6.3
CR _{2→1}	14/19	78.92	3.5
MI & CR _{1→2}	11/19	-64.45	5.95
MI & CR _{2→1}	14/19	76.28	3.85

Before merging CR and MI within a single value, both SMs need to be normalized in order for their values to be compared. As the CR measure already returns values between 0 and 1 we choose to linearly rescale MI within this range using its extreme estimated values.

Section VI will present matching results on real sonar images (Figs. 2–4), showing that this new MI&CR SM outperforms MI or CR measures used alone.

2) *Where Does the Formula Come From?:* When both random variables X and Y have a Gaussian distribution, a direct relationship between $CR(X, Y)$ and $MI(X, Y)$ exists [42] as given by

$$CR(X, Y) = \sqrt{1 - \exp(-2MI(X, Y))}. \quad (9)$$

In that specific Gaussian case, merging CR and MI into MI&CR new SM does not bring any extra information as they can be computed from each other. However, (9) gives us a way to put these two different measures CR and MI within a common description space.

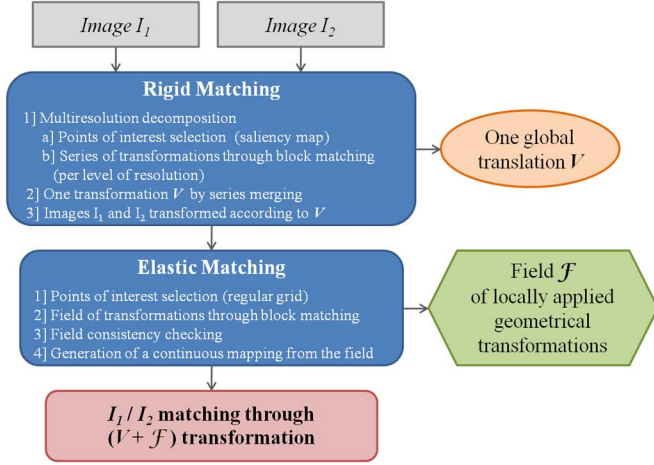


Fig. 6. Full registration algorithm overview.

3) *Why Is the Gaussian Case Not a Problem?*: As demonstrated in the previous paragraph, merging CR and MI values from two Gaussian-distributed blocks is useless since for such distribution laws, these quantities are linked by (9). The proposed algorithm is not affected by this case. Indeed, all the trials for matching blocks (and the subsequent similarity computations) are performed using blocks centered on image points of interest. As explained in Section IV, these points of interest correspond to maximum curvature estimations, i.e., locations where a scene feature affects the image through contours or texture changes. This implies that these blocks clearly do not have a Gaussian distribution but a more bimodal one. The new MI&CR SM is then fully relevant for the proposed matching algorithm that will work on these specific locations.

The next section will detail the full matching algorithm we propose to match sidescan sonar images.

V. PROPOSED REGISTRATION ALGORITHM

Based on the new SM MI&CR previously introduced, this section will present the complete registration algorithm that relies on this SM to find both global and local geometric transformations between the images to match. As shown in Fig. 6, the proposed algorithm is a two-step processing:

- a **rigid** matching phase computes a global mean geometric transformation between the images to be registered, through a multiresolution framework as shown in Fig. 9;
- then, using this global transformation as an initialization, an **elastic** matching procedure samples both images to compute a field of finer locally applied transformations (see Fig. 12).

Both matching phases rely on a common generic BLOCKMATCHING algorithm that is able to find the best matching in the test image I_2 , for a subimage from reference image I_1 .

A. Block Matching Algorithm

The proposed block matching algorithm consists in finding the best point in image I_2 that matches a given point p_1 from image I_1 , according to the MI&CR SM. To do so, as shown in

Fig. 7, the BLOCKMATCHING function iteratively searches for such a point in I_2 considering p_1 neighborhood (or block) and consistency criteria. Four steps are required to compute a robust match.

1) *Defining Neighborhood*: Working on p_1 neighborhood allows to take into account the whole set of image features around p_1 reducing speckle noise impact. Thus, according to Fig. 8, the NEIGHBOR function builds p_1 neighborhood $\{p_{2:5}\}$ as four extra points p_2 to p_5 , corners of a $w_{nei} \times w_{nei}$ window centered on p_1 with $w_{nei} = 39$.

Centered on each p_k point, a $w_{sim} \times w_{sim}$ subwindow with $w_{sim} = 17$ is considered when estimating SMs for p_k possible matching.

2) *Best Matching Point Search*: For each point p_k (from p_1 to p_5) in image I_1 , the FINDMATCHINGPOINT function searches for the best matching point in image I_2 . It consists in computing the SM between image I_1 blocks (p_k centered) and image I_2 blocks (m centered), over w_{sim} neighborhoods. m means all the locations in a subsampled version of image I_2 every $grid = w_{nei}/2$ pixels, to reduce computation time. Point m_k from I_2 having the highest SME s_k is then matched with point p_k from I_1 , creating a vector $v_k = m_k - p_k$ that defines an individual and local geometric transformation between the two images, with an associated s_k SME.

3) *Consistency Check for the Geometrical Transformation Field*: For each initial point p_1 , a field of five local geometric transformations $v_{1:5}$ associated with their SMEs $s_{1:5}$ is then found. The function CONSISTENCYCHECK1 checks whether this five-vector field exhibits direction and norm consistent behaviors. Thus, these vectors must check:

- an angular constraint: the five-vector directions must fit within a $\Delta\Theta^\circ$ angular sector;
- a norm constraint: the five-vector norms must also verify a consistency criterion

$$|||v_i|| - \bar{v}| \leq \nu \bar{v} \quad (10)$$

where v_i corresponds to one of the five individual transformations, $\bar{v}$ is the estimation of the five-vector norms mean and ν tunes the strength of the constraint.

These constraints mean that both directions and norms of the vectors are not too spread out. $\Delta\Theta$ and ν actual values that tune the spread tolerance, depend on the matching phase either rigid or elastic. If this field is too wide, p_1 is rejected as a candidate for matching. The next point in SME ordered list $\{p^{I_1}\}$ is then considered, starting over the loop.

4) *Final Geometric Transformation Computation*: If accepted, a final geometric transformation is built on the top of the five p_k points and their matching transformations. This final transformation $\bar{v}$, a pure translation, results from averaging the two vectors from $v_{1:5}$ having the highest SMEs. The corresponding global similarity measure estimate (GMSE) ψ is computed by

$$\psi = \frac{\max\{s_{1:5}\}}{\sigma} \quad (11)$$

BLOCKMATCHING (I_1, I_2, p_1)

- I_1 : Reference image (at level of resolution r)
 - I_2 : Test image (at level of resolution r)
 - p_1 : Point from I_1 to be matched in I_2
-

```

1: ▷ Anchor point  $p_1$  and its 4-neighbors  $p_{2:5}$  (see Fig. 8)
2:  $\{p_{2:5}\} \leftarrow \text{NEIGHBOR}(I_1, p_1, w_{nei})$ 
3:
4: ▷ Individual geometric transformations  $v_k$  with the corresponding SM  $s_k$  for the anchor point  $p_1$  and its
   neighbors  $p_{2:5}$ 
5: for all  $k \in 1 : 5$  do
6:    $[v_k, s_k] \leftarrow \text{FINDMATCHINGPOINT}(I_1, p_k, I_2, w_{sim})$ 
7: end for
8:  $\sigma \leftarrow \text{var}(s_{1:5})$ 
9: if  $\text{CONSISTENCYCHECK1}(v_{1:5})$  false then
10:  return  $\{\}$ 
11: end if
12:
13: ▷ Best individual geometric transformations
14:  $\{s_1^*, s_2^*\} \leftarrow \text{Two first maxima from } \{s_{1:5}\}$ 
15:  $\{v_1^*, v_2^*\} \leftarrow \{s_1^*, s_2^*\}$  corresponding transformations
16:
17: ▷ Final geometric transformation for matching  $p_1$ 
18:  $\bar{v} \leftarrow \text{mean}(v_1^*, v_2^*)$ 
19:  $\psi \leftarrow \max(s_1^*, s_2^*) / \sigma$ 
20:
21: return  $\{p_1, \bar{v}, \psi\}$  : geometric transformation  $\bar{v}$  matching point  $p_1$  from image  $I_1$  to image  $I_2$ , along with the
   corresponding SM  $\psi$ 

```

Fig. 7. Generic block matching algorithm between two images for one level of resolution.

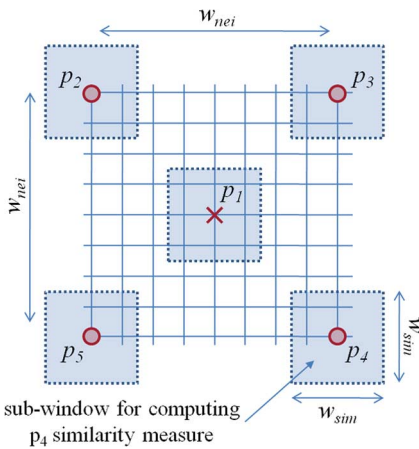


Fig. 8. Neighbors pattern for the block matching algorithm.

where $\max\{s_{1:5}\}$ is the maximum of the SMEs (over the five vectors) and σ , the $\{s_{1:5}\}$ standard deviation. Indeed, when the five SMEs, coming from the subwindow centered on the point of interest and the four subwindows centered at the four corners of the point of interest neighborhood, are close (σ is low), we

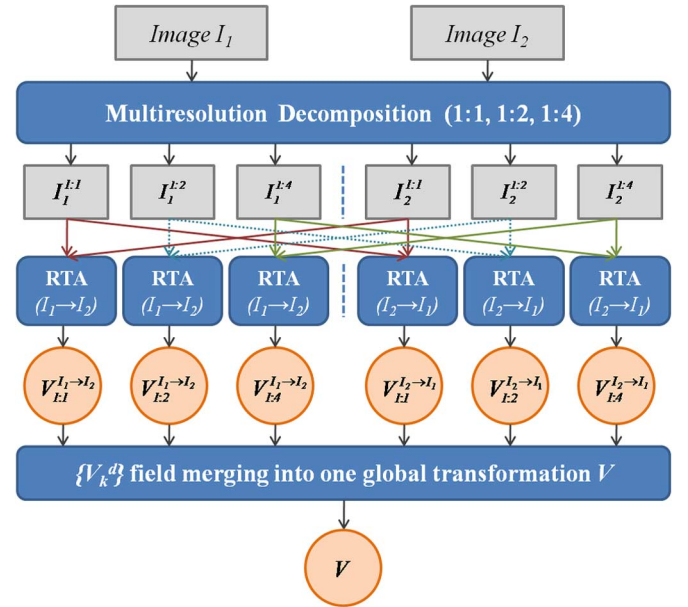


Fig. 9. Rigid matching step. RTA stands for rigid transformation algorithm and calls the RIGIDTRANSFORMATION function detailed in Fig. 10.

generally observe that a valid correspondence between the images has been found.

```

RIGIDTRANSFORMATION ( $I_1^{(r)}, I_2^{(r)}$ )
  •  $I_1^{(r)}$  : Reference image (resolution scale  $r$ )
  •  $I_2^{(r)}$  : Test image (resolution scale  $r$ )

```

```

1: ▷ Points of interest extraction
2:  $p^{I_1} \leftarrow$  Image  $I_1^{(r)}$  saliency points
3:
4: ▷ Series of geometric transformations  $\bar{v}_i$  matching saliency points from image  $I_1^{(r)}$  to image  $I_2^{(r)}$ 
5:  $\{p_i, \bar{v}_i, \psi_i\} \leftarrow \emptyset$ 
6: for all  $p_1 \in p^{I_1} \wedge \text{card}\{p_i, \bar{v}_i, \psi_i\} < 10$  do
7:    $\{p_i, \bar{v}_i, \psi_i\} \xleftarrow{+}$  MATCHINGBLOCK( $I_1^{(r)}, I_2^{(r)}, p_1$ )
8: end for
9:
10: ▷ Rigid transformation computation
11:  $\{\bar{v}_1^*, \bar{v}_2^*\} \leftarrow$  two vectors from  $\{\bar{v}_i\}$  corresponding to the two first maxima in  $\{\psi_i\}$  such that CONSISTENCY-
    CHECK2 ( $\bar{v}_1^*, \bar{v}_2^*$ ) is a success
12:  $V_r^{I_1 \rightarrow I_2} \leftarrow \text{mean}(\bar{v}_1^*, \bar{v}_2^*)$ 
13:
14: return  $V_r^{I_1 \rightarrow I_2}$  : One geometric transformation matching image  $I_1$  to image  $I_2$ , for level of resolution  $r$ 

```

Fig. 10. RIGIDTRANSFORMATION algorithm: computing a rigid geometric transformation matching two images, for one resolution scale.

B. Rigid Matching Step

The goal for the rigid matching step is to determine a global geometric transformation for wide areas in both images. Finer local geometric transformations will be computed starting from this global transformation, during the elastic matching phase.

As shown in Fig. 9, the rigid matching step mainly consists in determining one geometric transformation matching I_1 with I_2 and *vice versa*, for every resolution scale, through the RIGIDTRANSFORMATION algorithm (see Fig. 10). Only the best transformations according to their GSME are then used to compute the final rigid transformation matching images I_1 and I_2 instead of averaging the transforms from each resolution. In fact, by selecting the “best” transform, we select the transform based on the most relevant features.

Before using a multiresolution framework, a key feature of the proposed rigid matching step is to search for a geometric transformation between images only on points of interest. Indeed, focusing only on these relevant features or anchors permits to reduce the amount of individual estimated geometrical transformations embedded in the global rigid one.

1) *Finding a Rigid Transformation Per Resolution Scale:* The RIGIDTRANSFORMATION algorithm (see Fig. 10) explains how a single transformation matching one reference image to one test image is computed, mainly through the BLOCKMATCHING function applied to points of interest.

a) *Selecting points of interest:* Several methods for extracting points of interest have been developed in the past few

years [29], mainly for detecting relevant areas in images corresponding to human attention focusing. Indeed, these areas correspond to strong variations in the image and usually come from scene key features such as objects contours or homogeneous texture areas borders. Thus, these points are perfect candidates for being used as anchors for image registration processing.

Among the possible methods to extract anchor points [29], we chose the saliency map due to its noise robustness. It performs multiresolution images decomposition to detect relevant features based on both intensity and orientation image characteristics [43]. This map informs on variations of the image intensity contrast and responds to organized orientation within a pixel neighborhood. As the map is produced by dedicated processing (retinal model filter for contrast measurement and series of Gabor filters for orientation detection) at various scales, performing across scale image differences allow to adapt the feature detection mechanism to scene feature size (see [44] for more details). Finally, the saliency points that serve as points of interest correspond to pixel locations with the highest values within the saliency map. Comparing curvature and saliency points computed on the same sonar image reveals a more relevant localization of the saliency points around scene features, leaving behind homogeneous image areas [45]. Moreover, the saliency points better focus on areas where the image local histogram presents a strongly non-Gaussian behavior, involving the two proposed SMs. CR and MI cannot be linked using (8). Thus, these specific saliency points are ideally suited for the computation of an SM combining MI & CR as these

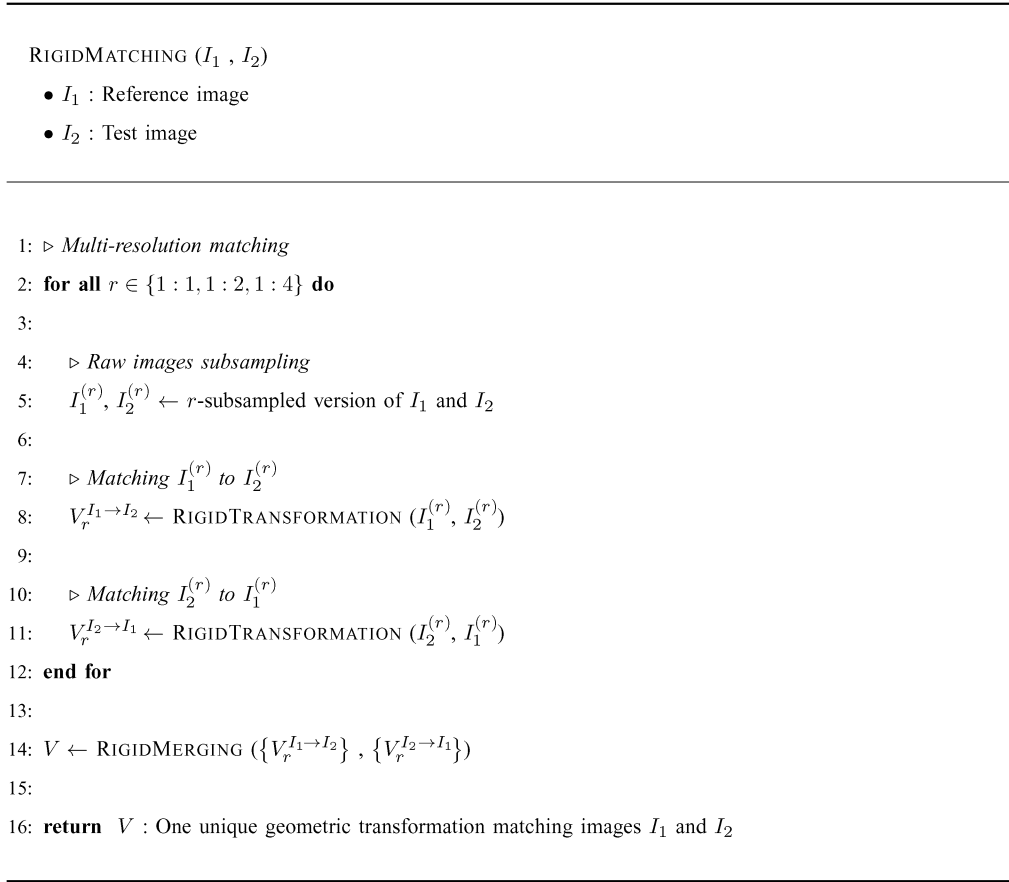


Fig. 11. Computing a unique geometric transformation matching images I_1 and I_2 through a symmetrical multiresolution rigid algorithm.

candidate points for registration take full benefit from both MI & CR behaviors.

b) Computing local transformations through block matching: As previously mentioned, in the rigid phase, the BLOCKMATCHING algorithm is fed by a list of image I_1 points of interests, sorted according to their score of saliency. Only the first ten points of interest respecting the consistency check CONSISTENCYCHECK1 with $\Delta\Theta = 90^\circ$ and $\nu = 0.8$ are considered as accurate candidates for computing a geometrical transformation between image I_1 and I_2 . Indeed, extra tests have shown that increasing the amount of local vectors did not improve the final rigid matching estimation.

c) A global rigid transformation per level of resolution: For each level of resolution, a series of ten triplets $\{p_i, \bar{v}_i, \psi_i\}$ has been obtained. Among these transformations $\{\bar{v}_i\}$, the two $\bar{v}_1^*$ and $\bar{v}_2^*$ having the highest GSME are averaged to produce one global transformation V_r matching image I_1 to image I_2 . However, to be selected, $\bar{v}_1^*$ and $\bar{v}_2^*$ must pass a new consistency check CONSISTENCYCHECK2 that will enforce a stronger norm consistency for the two transformations to be merged

$$\|\bar{v}_1^*\| - \|\bar{v}_2^*\| \leq \frac{1}{2} \max(\|\bar{v}_1^*\|, \|\bar{v}_2^*\|). \quad (12)$$

For each level of resolution r , a global transformation V_r is thus computed.

2) Working Through Multiresolution to Produce a Global Rigid Transformation: As shown in Fig. 11, the full rigid algorithm simultaneously processes images from three levels of resolution, using for each level the RIGIDTRANSFORMATION algorithm: full resolution (1:1), half resolution (1:2), and a quarter resolution (1:4). This multiresolution decomposition is achieved by a crude subsampling of the original images, so that no *a priori* hypothesis is made about the image contents.

Thus, with three levels of resolution, three transformations $V_r^{I_1 \rightarrow I_2}$ are computed for matching image I_1 to image I_2 . To produce a more robust final transformation, the same operation is performed for matching image I_2 to image I_1 . Indeed, the opposite transformation should be retrieved when switching the two images to match. Moreover, each transformation can be computed independently taking full benefit from distributed computer systems.

A final procedure RIGIDMERGING then generates transformation V globally matching images I_1 and I_2 , using only the transformations (among those obtained from the three resolutions) checking the two next criteria:

- opposite direction checking: within a level of resolution r , the two reverse transformations $V_r^{I_1 \rightarrow I_2}$ and $V_r^{I_2 \rightarrow I_1}$ must be as opposite as possible, thus verifying

$$\left| \left(V_r^{I_1 \rightarrow I_2}, \widehat{V_r^{I_2 \rightarrow I_1}} \right) - 180^\circ \right| < 90^\circ; \quad (13)$$

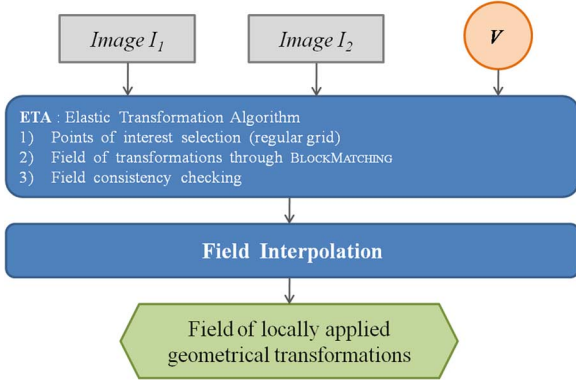


Fig. 12. Elastic matching step. ETA stands for the ELASTICMATCHING algorithm detailed in Fig. 13.

- norms consistency: within a level of resolution r , the two reverse transformations $V_r^{I_1 \rightarrow I_2}$ and $V_r^{I_2 \rightarrow I_1}$ must have similar lengths and check

$$\frac{1}{2} \|V_r^{I_1 \rightarrow I_2}\| \leq \|V_r^{I_2 \rightarrow I_1}\| \leq 2 \|V_r^{I_1 \rightarrow I_2}\|. \quad (14)$$

The final operation averages all the remaining consistent transformations taking into account their direction of matching, to produce a rigid global transformation V .

C. Elastic Matching Step

Once the rigid transformation has been computed and applied, the two images approximately match except for local remaining distortions. The elastic matching step consists in finely computing the field of these elementary distortions as shown in Fig. 12.

The key hypothesis is to consider these distortions as local rigid transformations because of their reduced influence range. Obviously, these local distortions can be computed through the BLOCKMATCHING algorithm operating all over image I_1 to build a field of locally applied transformations. The procedure ELASTICMATCHING detailed in Fig. 13 is performed on intermediate resolution images since both images have already been globally registered and accurate transformations are searched for.

1) *Gridding the Images for Field Computation:* As presented in Fig. 13, building a field of transformations over the full image starts with finding one local transformation matching every subimage $\hat{I}_{m,n}$ from image I_1 to image I_2 . Selecting the size Δ of these nonoverlapping subimages allows tuning both the description accuracy of the local distortions (vector field resolution) and the required processing time.

2) *Finding Local Transformations Through Block Matching:* To find the best local rigid transformation for every nonoverlapping subimages $\hat{I}_{m,n}$, a set of interest points $\{p^{I_{m,n}}\}$ is computed within every $\hat{I}_{m,n}$ and feeds the generic BLOCKMATCHING function (see Fig. 7 for more details). The best local triplet $(p, \bar{v}, \psi)$ corresponds to the point of interest having the highest saliency and checking the consistency constraints CONSISTENCYCHECK1 tailored for elastic matching:

- the angular constraint in which five vectors must fit in is loosened up to $\Delta\Theta = 120^\circ$ as the remaining trans-

formation can be small involving very small vectors with spreadout directions;

- the coefficient for the norm constraint in (10) is enforced to $\nu = 0.5$;
- for each point of interest p_1 , an extra constraint on the SMEs $s_{1:5}$ is added to the CONSISTENCYCHECK1: the five SMEs $s_{1:5}$ should stay in a predefined range. By ensuring rather homogeneous SMEs within the studied w_{nei} neighborhood ($p_{2:5}$) around the point of interest p_1 , this criterion eliminates points with isolated, very high SMEs. Indeed, it has been observed that the geometrical transformations induced by these points are mostly incorrect because they reflect occasional high SMEs that do not characterize image matching features relevant to the real scene.

Iterating over all the I_1 subimages $\hat{I}_{m,n}$, the locally applied transformations define a field of vectors $\{p_i, \bar{v}_i, \psi_i\}$.

3) *SMs Strength Checking:* Some nonrelevant transformations from the field $\{p_i, \bar{v}_i, \psi_i\}$ are induced by matching blocks having weak GSMEs. To remove these local false transformations, all the vectors v_i with a SME lower than a threshold are discarded through STRENGTHCHECKING function, according to

$$\psi_k \geq \text{mean}\{\psi_i\}/2. \quad (15)$$

4) *Continuous Mapping:* As the previous step generates a rather sparse vector field, interpolating between these vectors is required to get a consistent matching transformation for every pixel in the image. Inside the produced field, linear regression, thin plate spline, or even working with 1-D profiles computed on one image axis and extended to the full image can be used [8]. Outside the field boundaries, local linear extrapolations may be applied as each transformation operates within a local frame where the elastic transformations are represented by affine displacements.

VI. RESULTS

The proposed framework allows overlapping sonar tracks to be automatically registered and full sonar mosaic to be built from them. Indeed, this procedure needs same sea bottom areas within these tracks to be identified before any sonar image pixels merging. However, before presenting such results, this section will illustrate that coupling MI and CR into a new SM brings a more robust behavior for matching sonar images, compared to MI or CR SMs used separately.

For the sake of consistency, in the first part of this section, only results using set B are presented, results for sets A and C being close.

A. Coupling MI and CR Into a New SM: A Validation

1) *Experimental Data:* The results presented in this section are obtained using Fig. 3 sonar images. These images come from two slightly shifted tracks with opposite headings, on the same sea bottom area. Thus, one of the two images has been 180° rotated so that matching these two images only consists in finding an appropriate translation $(\Delta x, \Delta y)$. A visual estimation using selected interest points to evaluate an average displacement vector field $B_1 \rightarrow B_2$ between the two images B_1

```

ELASTICMATCHING ( $I_1, I_2$ )
  •  $I_1$  : Reference image
  •  $I_2$  : Test image

```

```

1: ▷ Series of locally applied rigid transformations  $\{p_i, \bar{v}_i, \psi_i\}$  from image  $I_1$  to image  $I_2$  (one per  $I_1$  sub-image)
2:  $\{p_i, \bar{v}_i, \psi_i\} \leftarrow \emptyset$ 
3: for all sub-image  $\hat{I}_{m,n} = I_1(m : m + \Delta, n : n + \Delta)$  do
4:
5:   ▷ Points of interest extraction within the sub-image  $\hat{I}_{m,n}$ 
6:    $\{p^{\hat{I}_{m,n}}\} \leftarrow$  saliency ordered list of sub-image  $\hat{I}_{m,n}$  points
7:   ▷ Building  $\{p_i, \bar{v}_i, \psi_i\}$  with all  $I_1$  sub-images analysis
8:   for all  $p \in \{p^{\hat{I}_{m,n}}\}$  do
9:
10:    ▷ One rigid transformation  $(p, \bar{v}, \psi)$  from sub-image  $\hat{I}_{m,n}$  to image  $I_2$ 
11:     $(p, \bar{v}, \psi) \leftarrow \text{MATCHINGBLOCK}(I_1, I_2, p)$ 
12:    if  $(p, \bar{v}, \psi) \neq \emptyset$  then
13:       $\{p_i, \bar{v}_i, \psi_i\} \leftarrow^+ (p, \bar{v}, \psi)$ 
14:      break
15:    end if
16:  end for
17: end for
18:
19: ▷ Similarity measures strength global checking
20:  $\{M_k, \bar{v}_k, \psi_k\} \leftarrow \text{STRENGTHCHECKING}(\{p_i, \bar{v}_i, \psi_i\})$ 
21:
22: return  $\{M_k, \bar{v}_k\}$  : Field of locally applied geometric transformations, matching image  $I_1$  to image  $I_2$ 

```

Fig. 13. Elastic matching algorithm.

and B_2 brings $(\Delta x, \Delta y) = (-65, -4)$ pixels to be considered as the transformation to be found by our algorithm.

2) *Evaluation Methodology*: To evaluate the benefits of coupling MI and CR, a series of matching has been conducted between images B_1 and B_2 using three different SMs: MI alone, CR alone, and coupled MI&CR. As previously described, points of interest from B_2 have been taken as anchor points for computing these SMs. In this test, we chose to compute only 19 transformations corresponding to the 19 best extracted saliency points as we observed a drop of the saliency score for the subsequent points. Moreover, as images B_1 and B_2 can be registered through a unique transformation, the 19 computed transformations should be similar and close to $(\Delta x, \Delta y)$.

Thus, a criterion β has been defined to compare the results output by the three SMs, as the ratio of *valid* to total computed transformations (here, 19). β basically informs about the consistency of the transformation field found by various SMEs, according to the effective transformation. Indeed, a *valid* transformation must pass all the checks described in Fig. 7 BLOCKMATCHING algorithm, especially those analyzing

the consistency of the computed transformations over a local neighborhood. A β value close to 1 indicates a reliable transformation field.

Coming with each β value, a global transformation $(\Delta x, \Delta y)$ is computed by averaging all the nonrejected individual transformations.

3) *Results*: Table II presents the results of the evaluation procedure on B_1 and B_2 images, for three levels of resolution. The 256×256 resolution corresponds to the full image resolution while 64×64 resolution is related to 1:4 sampled images. $\text{SM}_{a \rightarrow b}$ denotes that the SM has been used for matching image a to image b ; in that case, image a is considered as the *reference* image, and image b as the *test* image.

MI-alone measure produces poor results ($\beta = 1/19$ and $\beta = 0/19$) for the lowest resolution: no main transformation can be distinguished. Stronger β values are obtained ($\beta = 11/19$ and $\beta = 14/19$) for the highest resolutions, confirming that data undersampling weakens MI relevancy.

On the contrary, CR-alone measure shows satisfying results for lower resolutions (from $\beta = 3/19$ to $\beta = 8/19$). Indeed,

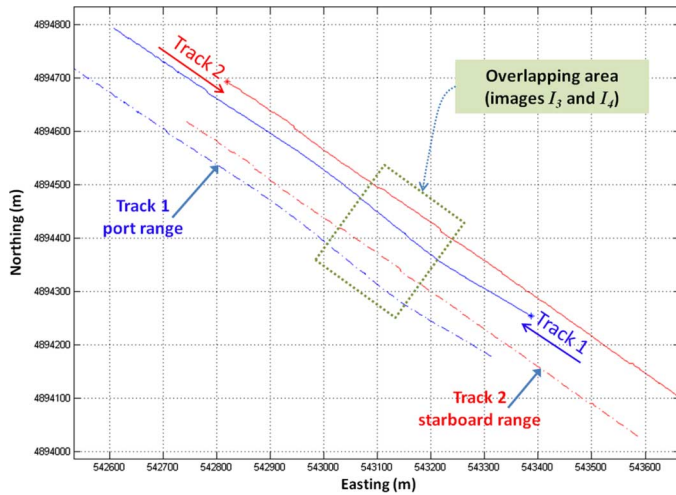


Fig. 14. Raw ship location recorded along tracks 1 and 2.

CR being based on variance estimation on a local neighborhood is less influenced when reducing the resolution.

Finally, it appears that the MI&CR combined measure generates the most reliable matching measure, with higher β values for each resolution level. Moreover, coming with β high values and thus consistent individual geometrical transformations, the average transformation clearly gives a good approximation of the effective transformation, over the resolution levels.

In addition, Table II illustrates the impact of selecting the reference image among the two images to match. Indeed, the nonsymmetrical behavior of the β criterion is clearly pointed out as in our case, when image B_2 is selected as a reference image, β values are higher. This is consistent with the observation of image B_2 which contains fewer shadows than image B_1 ; as a consequence, there is more information sources on image B_2 making the matching easier.

This new SM MI&CR will then be used in the next section for full track registration.

B. Sonar Track Registration

To test the complete proposed algorithm, two overlapping tracks coming from at-sea acquisitions in the Ligurian sea (Italy) and displayed in Fig. 15 are considered. Previous sets B and C were extracted from these 4800×2276 pixel tracks, thus acquisition characteristics given in Section II-B1 still apply. Moreover, besides the correction (TVG performed by the sonar system before outputting the images) and the subsampling, no extra processing is run on these raw images before our registration algorithm. Actually the brighter across-track band appearing on the images indicates that some corrections remain to be performed on raw data, to get a perfect uniform backscattering along range.

Being part of a global survey, raw tracks 1 and 2 are slightly shifted with opposite headings, partially observing the same sea bottom area as indicated in Fig. 14, which presents the ship trajectory during these tracks. No towfish location has been recorded during the survey. We can only assume that the towfish globally follows the ship but its actual locations remain precisely unknown as the layback is around 80 m.

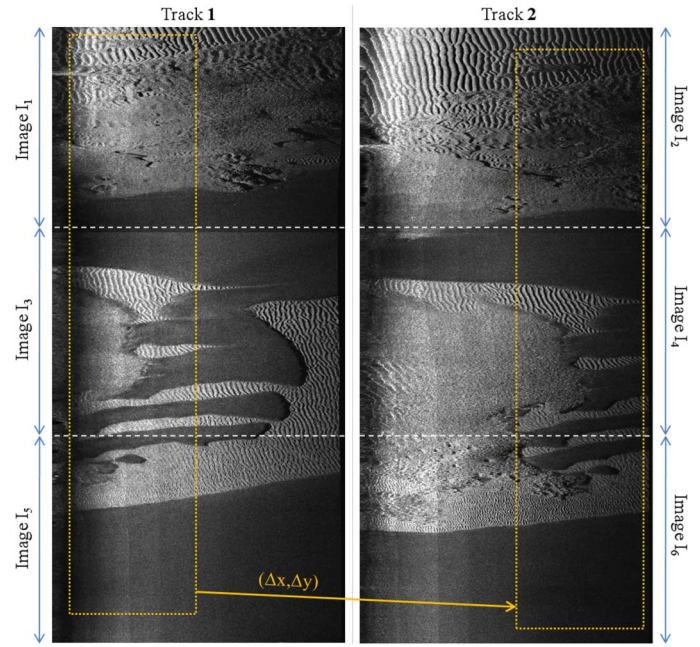


Fig. 15. Tracks 1 and 2 registration using subimages rigid matching.

The two tracks have opposite headings, thus track 2 has been 180° rotated such that matching them only consists in finding appropriate translations $(\Delta x, \Delta y)$ between track 1 port images and track 2 starboard images.

As shown in Fig. 15, each track is split in several equivalent parts (here, three images) to take into account small heading changes along the tracks and reduce the computing time required for matching images. Thus, track 1 (resp., track 2) is split into 400 pings images I_1, I_3 , and I_5 (resp., images I_2, I_4 , and I_6). Each image consists of 1600 lines as this Klein sidescan produces four beams per ping. This splitting allows matching each pair (I_k, I_{k+1}) of images to deal locally with unknown towfish location due to layback behavior which is the main reason for registration procedure.

A visual estimation using selected interest points to evaluate an average displacement vector between the two tracks brings the translation $(\Delta x, \Delta y) = (+973, -97)$ pixels to be considered as the rigid transformation to be found by our algorithm to match track 1 to track 2. It seems that for this particular application this rigid transformation is predominant due to the straight constant speed towfish displacement during the survey.

1) *Images Registration:* To perform full tracks registration for building the complete mosaic in the next step, the proposed algorithm independently matches images coming from each track (I_1 with I_2 , I_3 with I_4 , and finally, I_5 with I_6).

Tables III–V give the rigid geometrical transformation computed for each pair of images and for each level of resolution, along with their corresponding MI measures. As already mentioned and illustrated in these tables, MI measures are not reliable and cannot be used alone to select adequate transformations from this multiresolution framework to compute a global transformation. That was the reason why extra criteria (opposite directions and similar norms) had to be fulfilled by a geometrical transformation to be taken into account for

TABLE III
REGISTRATION IMAGES I_1 AND I_2 THROUGH A $(\Delta x, \Delta y)$ TRANSLATION ONLY, FOR SEVERAL LEVELS OF RESOLUTION

Matching	Resolution	Δx	Δy	MI
$I_1 \rightarrow I_2$	71×50	-408	184	0.4891
	142×100	-1224	-536	0.2634
	284×200	-220	-1016	0.4503
$I_2 \rightarrow I_1$	71×50	-512	176	0.4913
	142×100	-896	248	0.3287
	284×200	-904	180	0.3329
$R_{1 \rightarrow 2}$		900	-214	

TABLE IV
REGISTRATION IMAGES I_3 AND I_4 THROUGH A $(\Delta x, \Delta y)$ TRANSLATION ONLY, FOR SEVERAL LEVELS OF RESOLUTION

Matching	Resolution	Δx	Δy	MI
$I_3 \rightarrow I_4$	71×50	624	-96	0.4055
	142×100	1168	-72	0.3488
	284×200	796	-80	0.3406
$I_4 \rightarrow I_3$	71×50	256	336	0.3651
	142×100	-656	144	0.4003
	284×200	-1604	-676	0.2695
$R_{3 \rightarrow 4}$		873	-98	

TABLE V
REGISTRATION IMAGES I_5 AND I_6 THROUGH A $(\Delta x, \Delta y)$ TRANSLATION ONLY, FOR SEVERAL LEVELS OF RESOLUTION

Matching	Resolution	Δx	Δy	MI
$I_5 \rightarrow I_6$	71×50	560	-64	0.4676
	142×100	816	-136	0.4163
	284×200	630	-172	0.5893
$I_6 \rightarrow I_5$	71×50	608	304	0.7623
	142×100	728	320	0.7244
	284×200	-708	140	0.4549
$R_{5 \rightarrow 6}$		718	-149	

computing the global resulting transformation. Only vectors written in bold check all the criteria. Thus, for each pair of images, these successful vectors are averaged to give final transformations $R_{1 \rightarrow 2}$, $R_{3 \rightarrow 4}$, and $R_{5 \rightarrow 6}$, appearing in the last row of each table.

The computing time depends on the images to match and stays within a 1:5 ratio. For image pair I_3 and I_4 (2276×1600 pixels), a 4'30'' processing time is necessary to evaluate the transformation on the two first levels of resolution, while requiring 22' for the three levels, with a Pentium IV processor.

2) *From Image to Track Matching*: On this specific set of data, the survey was performed with constant towfish speed along a straight path. Thus, the global transformation along the track Δy is connected to the towfish speed, and we consider that

it can be defined as the individual transformations average along the $\vec{y}$ axis

$$\Delta y = \frac{\Delta y(R_{1 \rightarrow 2}) + \Delta y(R_{3 \rightarrow 4}) + \Delta y(R_{5 \rightarrow 6})}{3}. \quad (16)$$

This equation gives $\Delta y = -153$ for the global translation along the reference track, allowing track 1 to be matched with track 2.

About the across-track direction, Δx is defined by a spline interpolation using the points of the three x -coordinates 900, 873, and 718 (as given in the last row of each table). Obviously, this matching step compensates mainly for consistent towfish navigation changes around its heading. The displacement vector field is then divided by two and applied symmetrically to track 1 and track 2.

When looking at Fig. 14, the average distance between the two tracks is between 30 and 35 m. Considering the 3.75-cm across-track pixel resolution and the 873-pixel shift of the rigid translation, the estimated rigid shift would be around 33 m, which is globally consistent with the ship recorded trajectory. This computation is just a very rough check as we are not using the effective towfish location but the ship location (assuming an ideal layback behavior) and we are directly comparing pixel resolution with distance on the sea bed. The last point introduces small variations as the estimated altitude of the towfish is around 10–15 m, for a 75-m range and midrange sonar data are used.

Remark: Fully georeferenced tracks are more difficult to match than raw tracks because of information losses caused by the georeferencing processing (filtering, smoothing, interpolation) as studied in [22]. Indeed, such a processing requires two interpolations: one along track to fill the lack of sea bottom coverage between consecutive pings, and one equivalent to a 1-D median filter, to fill the *holes* caused by the sonar data projection on the georeferenced grid.

3) *Elastic Step*: Fig. 19 shows the mosaic built from tracks 1 and 2 matched according to a rigid matching alone. We consider that this first rigid matching result could already constitute a satisfying mosaic. Consequently, our interest for the elastic matching case will focus on the pair of images I_3 and I_4 .

Fig. 16 presents the matching results using an elastic transformation on images I_3 and I_4 . For this experiment, the elastic algorithm parameters described in Fig. 13 are $w_{nei} = 49$ and $w_{sim} = 17$ with a $\Delta = 40$ pixels grid step. As shown in Fig. 16(a), the computed vector field is rather homogeneous, starting from no transformation at the top of the area to stronger displacements towards the bottom of the image.

Towfish displacements are mainly characterized by translations as previously mentioned. The previous rigid matching globally compensates for layback along $\vec{y}$ -axis and shifted trajectories along $\vec{x}$ -axis. The elastic procedure consists in computing the local remaining corrections to be applied for matching images I_3 and I_4 . These locally applied translations mostly compensate for local lateral displacement of the towfish during the survey. This explains the geometry of the elastic vector field, which is mainly characterized by variable length translations along $\vec{x}$ -axis [see Fig. 16(a)].

The elastic field behavior seems consistent with the ship location recorded during the survey, for the two tracks. Indeed, Fig. 17 zooms in on the area where tracks 1 and 2 probably

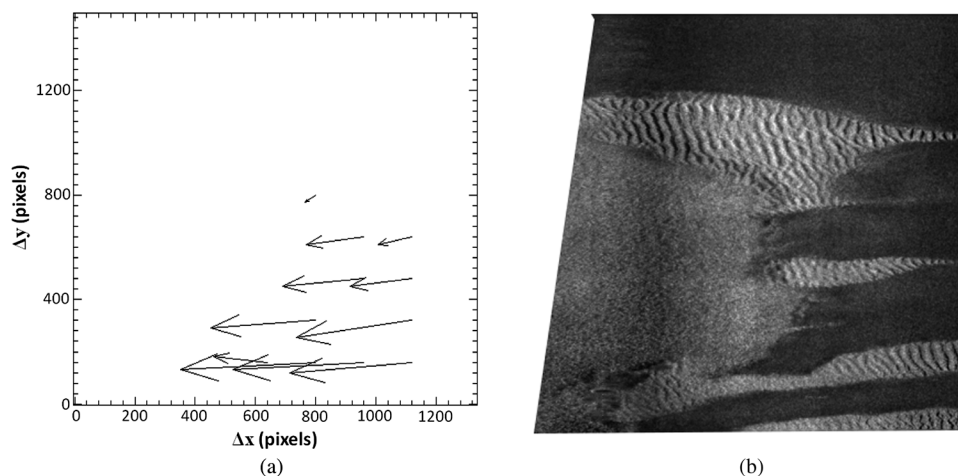


Fig. 16. Elastic matching on images I_3 and I_4 overlapping area. Images I_3 and I_4 have already been roughly registered through previously computed rigid transformation. (a) Field of translations allowing images I_3 and I_4 to match (zoom in on their overlapping area); (b) warped image I_4 after elastic registration.

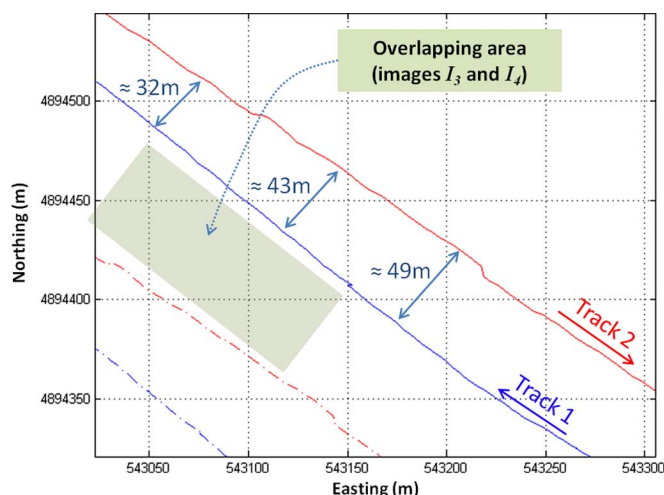


Fig. 17. Overlapping seabed area for images I_3 and I_4 according to raw ship location and approximative 75-m layback.

overlap over images I_3 and I_4 . It clearly shows that locally the two tracks move away from each other, starting from a global 32-m to a final 49-m translation before getting closer. When looking at the elastic field, we observe that the estimated global rigid procedure took into account the 32-m global shift as no local transformation is found at the top of the area. Then, as we move towards the bottom of the area, the translation length keeps growing to a final 400-pixel length, which represents around 15 m. Again, this roughly estimated shift is consistent with the distance between tracks at images I_3 and I_4 ends.

These computations are by no means a validation of the algorithm performances. However, it shows its consistent behavior when operating on real data.

Applying the estimated field of translations to image I_4 to match image I_3 produces a warped version of this image as shown in Fig. 16(b). Finally, the computed field has been used on the complete raw images to produce the full mosaic shown in Fig. 18.

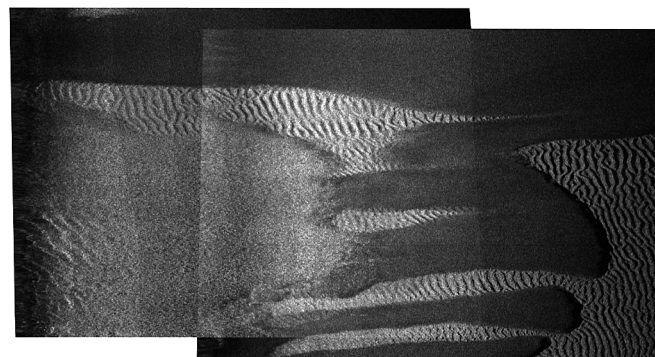


Fig. 18. Sonar image mosaic computed from images I_3 and I_4 after the full registration process (rigid and elastic).

4) *Mosaic Building*: Using an averaging policy for pixel merging, the sonar mosaic resulting from the previous registration step is presented in Fig. 19. Once the two tracks are matched, we only need a few reference points from the images with known latitude and longitude to achieve the full tracks (and thus mosaic) georeferencing. In this study, WGS84 geoid and UTM projection system have been used. The resolution of the mosaic pixels is about $3.3 \text{ cm} \times 3.3 \text{ cm}$, which allows a high precision for the texture analysis of the seafloor. It is remarkable that from the bottom to the top of the mosaic, the different textures add up very well, which means the matching estimation is well performed to get a satisfying seafloor imaging tool.

Remark: Where wide homogeneous areas appear, such as in images I_5 and I_6 , correct displacements have also been estimated. On such areas, this shows that iconic methods are well adapted compared to symbolic methods that fail.

VII. DISCUSSION

As already mentioned, any would-be general registration algorithm should deal with the three kinds of variabilities (spatial, temporal, and modal) encountered when getting multiple acquisitions from the same seabed area.



Fig. 19. Final sonar image mosaic ($30\,000 \times 25\,000$ pixels covering a 1×0.8 km² area).

In this paper, only topics related to *spatial* variability have been taken into account, within a well-defined scope of work, to get interesting registration results. Indeed, as we suppose sonar images are acquired during the same survey, we assume a specific pattern for the track trajectories involving parallel observations, constant altitude and speed, and an overlapping representing roughly 40% of the track width. These constraints ensure decent observation redundancy and minimum image scale variations between tracks. However, for practical and tactical reasons, it should be feasible to relax some of these constraints and our framework should be able to deal with the consequences: no possible matching or extra spatial variations.

As our framework checks and enforces the consistency of elementary computed transformations to produce either rigid or elastic matching, registration images that do not represent the same seabed could be rather easily detected. Indeed, when looking at the same area, fields of computed transformations are both across scale and locally consistent. Local consistency indicates that a feature from image 1 looks like another feature from image 2 while across scale takes into account how features are distributed in the observed scene. The current consistency scores can be extended to fully deal with variable overlapping.

Concerning extra spatial variations, mainly scaling, MI and CR SMs are supposed to be robust to image distortions. Moreover, performing across-scale matching instead of same-scale matching only could also deal with such image scale distortions. This technique could also be used for *multimodal* matching when images from two different imaging sonars must be merged within a unique mosaic. Indeed, sonar resolution differences could be compensated for by such an across-scale matching procedure.

About *temporal* variability, the proposed framework should be able to deal with time-distant surveys. However, some restrictions clearly apply as iconic matching is intensity based, which corresponds to low-level information. As long as images to be matched present the same features with close aspect, matching could be considered, especially to produce one

unique mosaic. However, if the physical properties or the configuration of the observed area change, involving drastic modifications in the collected images, iconic matching cannot cope with such variations. For instance, while staying at the same location, ripple fields can change their frequency and orientation over time. Symbolic matching after segmentation and classification operates on a higher informative level with attributes labeling the considered features (here, the *ripple field* label). Thus, such symbolic matching may be able to reference the same phenomenon (the ripple area) on various tracks, even if the phenomenon basic properties (its orientation or frequency) change. Iconic methods heavily relying on image intensity characteristics probably will not exhibit a consistent behavior. In any case, the problem will then be to produce a unique and useful mosaic embedding of such different viewpoints.

VIII. CONCLUSION

One of the major issues in sidescan sonar image processing is to automatically generate a mosaic. Many studies have presented solutions to this problem with symbolic matching approaches. The originality of this paper is to propose a framework using iconic methods to perform automatic sonar images matching, even on homogeneous areas where symbolic techniques fail due to the lack of symbolic features.

The challenge has been to integrate survey *a priori* knowledge and deal with the variations and uncertainties that any real experience brings. Thus, under some acceptable assumptions such as a flat seabed (no elevation information is produced by sidescan sensors) and small scale variations between tracks, we develop a two-step intensity-based block matching algorithm. First step roughly registers two sonar images through a single rigid transformation, mainly taking into account global shifts and towfish layback. Once the transformation has been applied to the images, the second step computes a field of locally applied translations to find the remaining small distortions between the images. This elastic matching guarantees local phenomena such as yaw or local shifts, to be considered.

Several innovations characterize the proposed intensity-based registration algorithm. First, we tuned the block matching mechanism to focus and operate within the neighborhoods of points of interest corresponding to maxima of saliency. Indeed, a relevant pruning of the matching possibilities clearly reduces computation cost compared to a blind block matching. Then, image blocks are compared through a stochastic approach involving the definition of a relevant SM. To do so, we designed a new SM MI&CR that embeds both MI and CR behaviors. More than 30 SMs including MI and CR have been benchmarked on real sonar images to select MI and CR for coupling. As MI and CR alone characterize different natures of a relationship; at different scales, between the compared images, the new SM takes full benefit from the proposed multiresolution framework and outperforms MI or CR working alone. Moreover, image locations where MI&CR takes profit from MI and CR synergy correspond to maxima of saliency. These pixels are precisely those where the block matching operates, pointing at the overall consistency of our framework.

The proposed algorithm has been applied to two tracks extracted from a real survey. The rigid step clearly grabs

global layback and shift between the two tracks, while the finer elastic step successfully integrates local heading variations. Both rigid and elastic steps produce relevant results and behave consistently when roughly compared to recorded ship trajectories. Matched images and subsequent mosaics have been then successfully produced.

Thus, within the discussion framework, the proposed method gives satisfying results for automatically generating sonar mosaics from nonregistered tracks.

ACKNOWLEDGMENT

The authors would like to thank the Groupe d'Etudes Sous-Marines de l'Atlantique (GESMA) for providing sonar data sets.

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of the Ushant front.

Cyril Chailloux received the Ph.D. degree in signal processing and telecommunications from the University of Rennes I, Rennes, France, in 2007, focusing on SONAR image registration.

From 2007 to 2008, he worked for the Ifremer Institute as a Postdoctoral Researcher on automatic classification of marine habitats from texture analysis. Currently, he is a Researcher for the Ecole Nationale Supérieure des Ingénieurs des Etudes et Techniques d'Armement (ENSTA Bretagne) in passive acoustics to analyze the dynamical structure



Jean-Marc Le Caillec (M'00–S'07) was born in 1968 in Paris, France. He received the Degree of Engineer in telecommunications from the Ecole Nationale Supérieure des Télécommunications de Bretagne (ENST Bretagne), Brest, France, in 1992 and the Ph.D. degree in mathematics and signal processing from the University of Rennes I, Rennes, France, in 1992.

From 1997 to 1999, he worked for Thomson AirSys (now Thales Air Defense) on Unsupervised Radar Management and he joined the TELECOM

Bretagne in 1999 where he is now Assistant Professor in Mathematics, Stochastic Processes, and Signal Processing. His main research interests are statistics, nonlinear system modeling, mathematics, and signal processing for applications in remote sensing of the ocean.



Didier Guériot (M'95–S'08) graduated from the Institut National des Sciences Appliquées (INSA), Rennes, France, in 1991, where he studied computer science. He received the Ph.D. degree in electronics from the University of Haute Alsace, Mulhouse, France, in 1998, focusing on extending genetic algorithms to variable length solutions for difficult problems such as neural networks learning and symbolic sonar image registration.

From 1993 to 2005, he was an R&D Engineer for several research labs and companies within the sonar field, dealing with high-resolution sonar imagery for Groupe d'Etudes Sous-Marines de l'Atlantique (GESMA, a French Navy research agency) and multi-beam echosounders for RESON in Santa Barbara, CA. In 2005, he joined Ecole Nationale Supérieure des Télécommunications (TELECOM Bretagne), Brest, France, as an Associate Professor. His research interests range from sonar data processing to artificial intelligence applied to underwater environment reconstruction.



Benoît Zerr (M'94) was born in Belfort, France, on September 20, 1961. He received the Ph.D. degree in electrical engineering from the University of Haute Alsace, Mulhouse, France, in 1989.

From 1986 to 1994, he was a Research and Development Engineer at Groupe d'Etudes Sous-Marines de l'Atlantique (GESMA), Brest, France. From 1995 to 2001, he was a Senior Scientist in the Mine Counter Measure group at SACLANT Undersea Research Centre (now called NURC), La Spezia, Italy. From 2001 to 2008, he was back at GESMA, occupying successively the positions of Head of High Frequency Acoustics Department, Scientific Advisor and National Expert in Mine Counter Measure Sonar Systems. Since 2003, he has been an Associate Researcher at Telecom Bretagne, Brest, France. In 2009, he was detached from French MOD to an academic position at ENSIETA postgraduate school, teaching electronics and underwater acoustics, and managing the sonar and underwater sensing research team of E312-EA3876 laboratory.